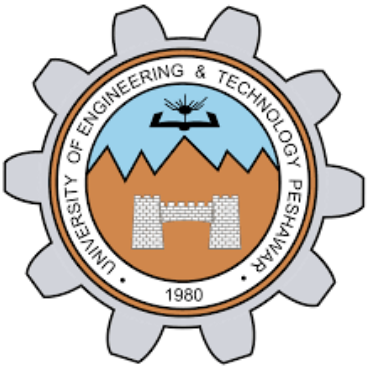
# CA Lab Project

# Bluetooth Home Automation System Using Arduino




CSE201L Circuit Analysis Lab
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# 1 Objectives:

- To design and implement a Bluetooth-based home automation system using Arduino UNO and the HC-05 Bluetooth module.
- To apply fundamental circuit analysis concepts such as Kirchhoff's Voltage Law (KVL), Kirchhoff's Current Law (KCL), relay switching, and load control in a practical hardware project.
- To establish wireless communication between a smartphone and Arduino UNO for remote control of electrical loads.
- To control four AC loads independently using a 4-channel relay module.
- To verify the functionality of the system through simulation and hardware implementation.

# 2 Components / Tools Required:

| S.No. | Component | Specification / Value |
|---|---|---|
| 1 | Arduino UNO | ATmega328P Microcontroller Board |
| 2 | HC-05 Bluetooth Module | Bluetooth Serial Communication Module |
| 3 | 4-Channel Relay Module | 5V Relay Board, 4 Independent Channels |
| 4 | AC Bulbs | Four 220V AC Lamps (Loads) |
| 5 | AC Power Supply | 220V AC Mains Supply |
| 6 | DC Power Adapter | Arduino Power Supply Adapter |
| 7 | Smartphone | Android Device with Bluetooth Control App |
| 8 | Connecting Wires | Jumper Wires for Interconnections |
| 9 | USB Programming Cable | Arduino Programming and Power Cable |

Table 1: Components and Tools Used in Bluetooth Home Automation System

# 3 Circuit Analysis Concepts Applied:

## 3.1 Kirchhoff's Voltage Law (KVL):

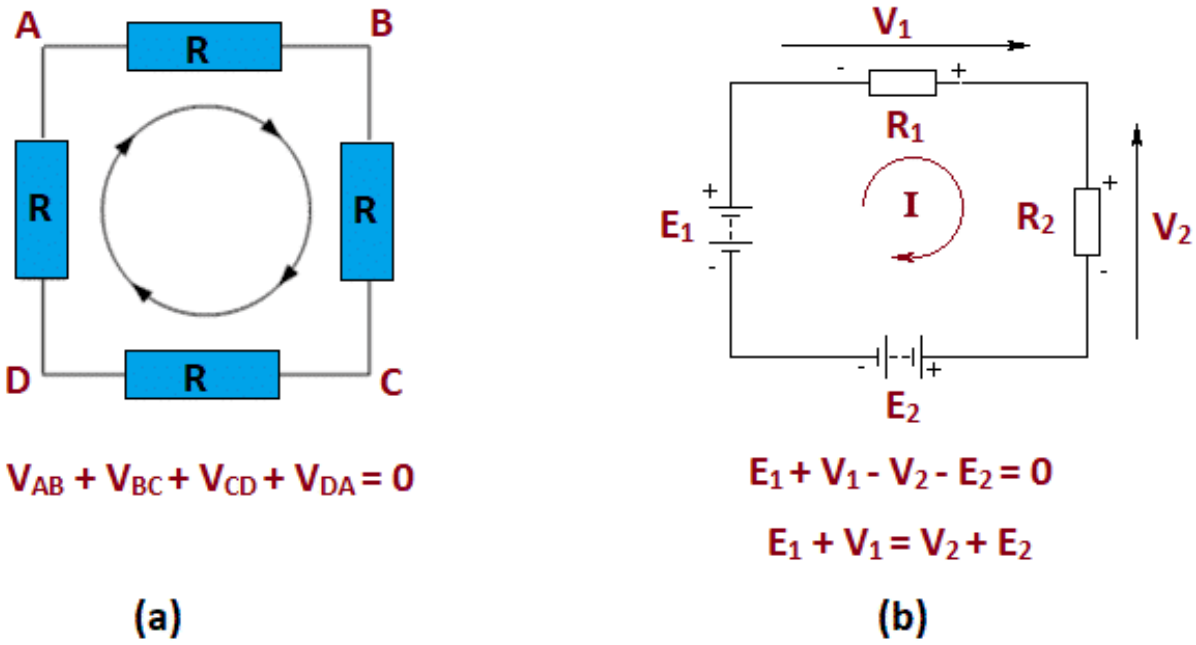



Figure 1: Kirchoff's Voltage Law

Kirchhoff's Voltage Law states that the algebraic sum of all voltages around a closed loop is zero.

$$\sum V = 0$$

For the Arduino power loop:

$$V_{supply} - V_{Arduino} = 0$$

$$5V - 5V = 0$$

This verifies that the supplied voltage is properly distributed to the Arduino and connected modules.

## 3.2 Kirchhoff's Current Law (KCL):

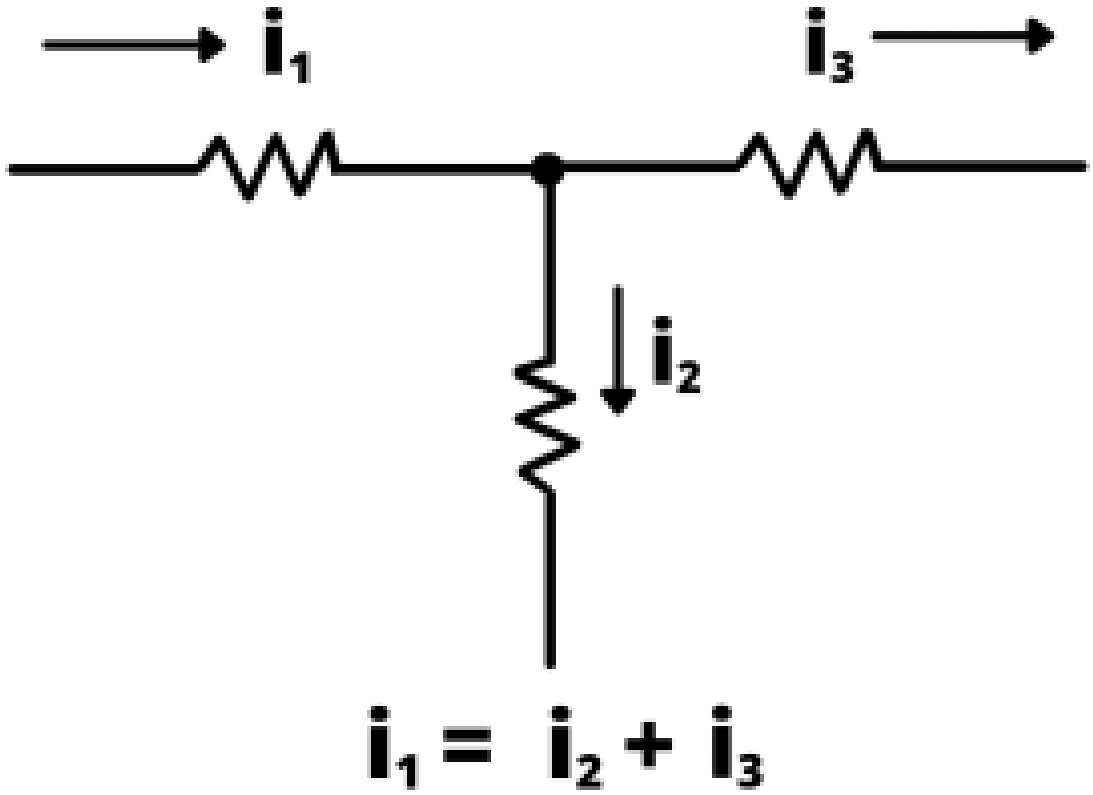



Figure 2: Kirchoff's Current Law(KCL)

Kirchhoff's Current Law states that the sum of currents entering a node equals the sum of currents leaving that node.

$$\sum I_{in} = \sum I_{out}$$

At the relay module power node:

$$I_{Arduino} = I_{Relay1} + I_{Relay2} + I_{Relay3} + I_{Relay4}$$

This confirms that the total current supplied by the source is distributed among the active relay channels.

## 3.3 Bluetooth Communication Analysis:

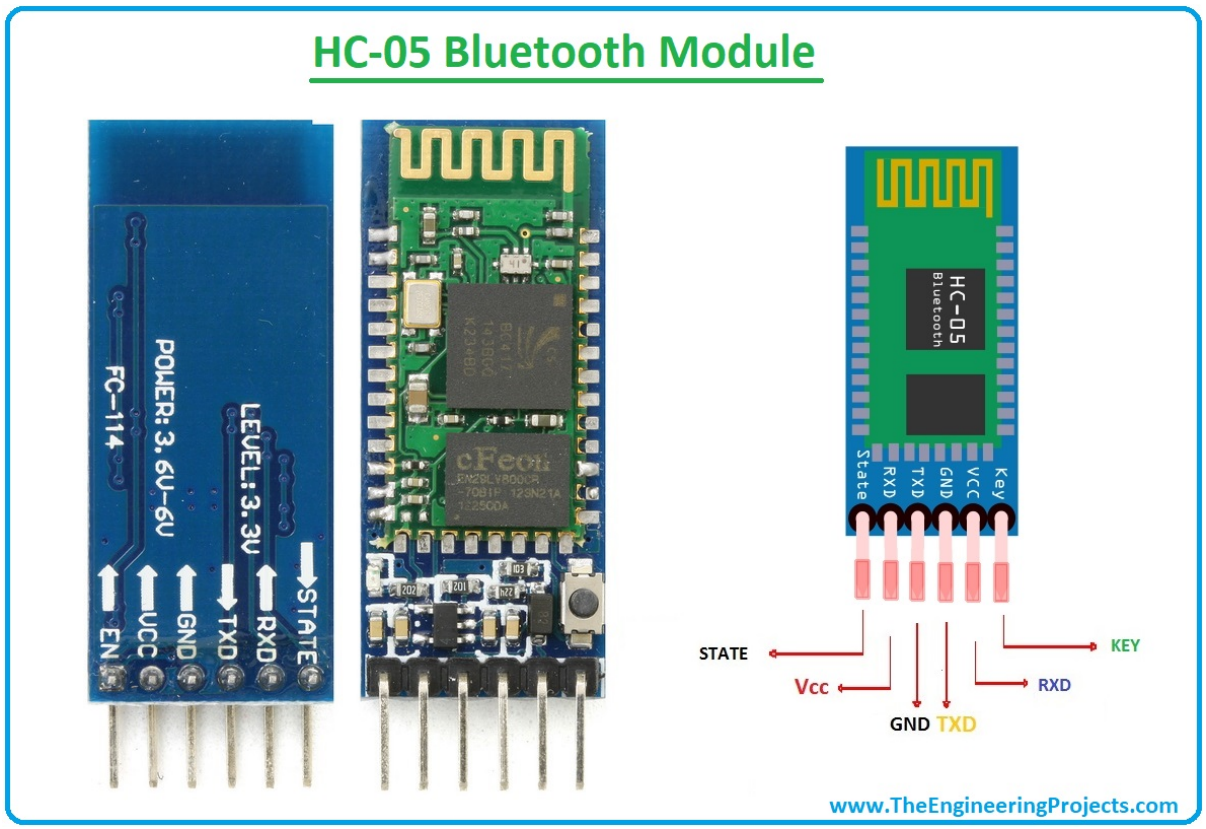



Figure 3: HC-05 Bluetooth Module

The HC-05 Bluetooth module establishes wireless serial communication between the smartphone and Arduino UNO.

The module operates at a baud rate of:

$$9600 \text{ bps}$$

The received Bluetooth commands are transmitted serially to the Arduino, which processes the commands and controls the corresponding relay channels.

## 3.4 Relay Switching Principle:

The 4-channel relay module acts as an electrically controlled switch.

When an Arduino output pin becomes HIGH:

$$V_{IN} = 5V$$

the corresponding relay is energized and the AC load circuit is completed.

When the Arduino output pin becomes LOW:

$$V_{IN} = 0V$$

the relay is de-energized and the AC load is disconnected.

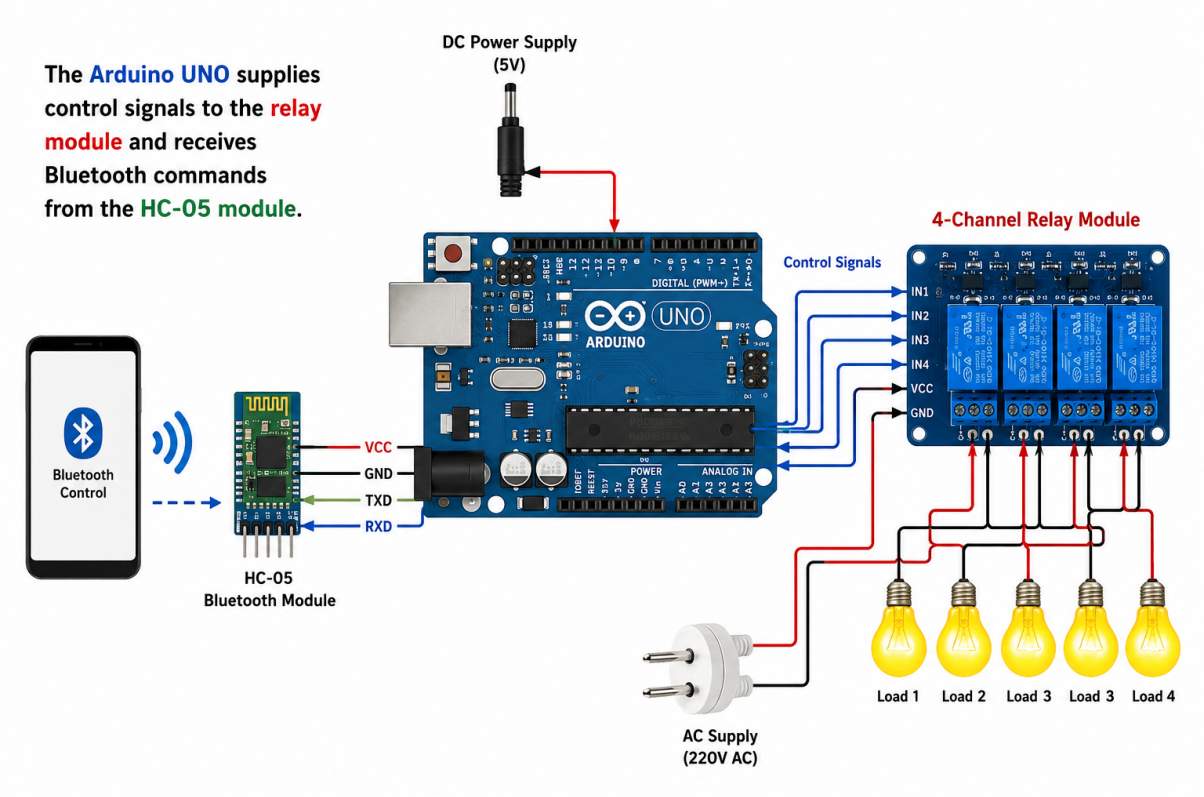



Figure 4: Bluetooth Home Automation System Architecture

## 3.5 Power Analysis:

The Arduino UNO supplies control signals to the relay module and receives Bluetooth commands from the HC-05 module.

The operating voltages are:

$$V_{Arduino} = 5V$$

$$V_{HC05} = 5V$$

$$V_{Relay} = 5V$$

$$V_{Load} = 220V\ AC$$

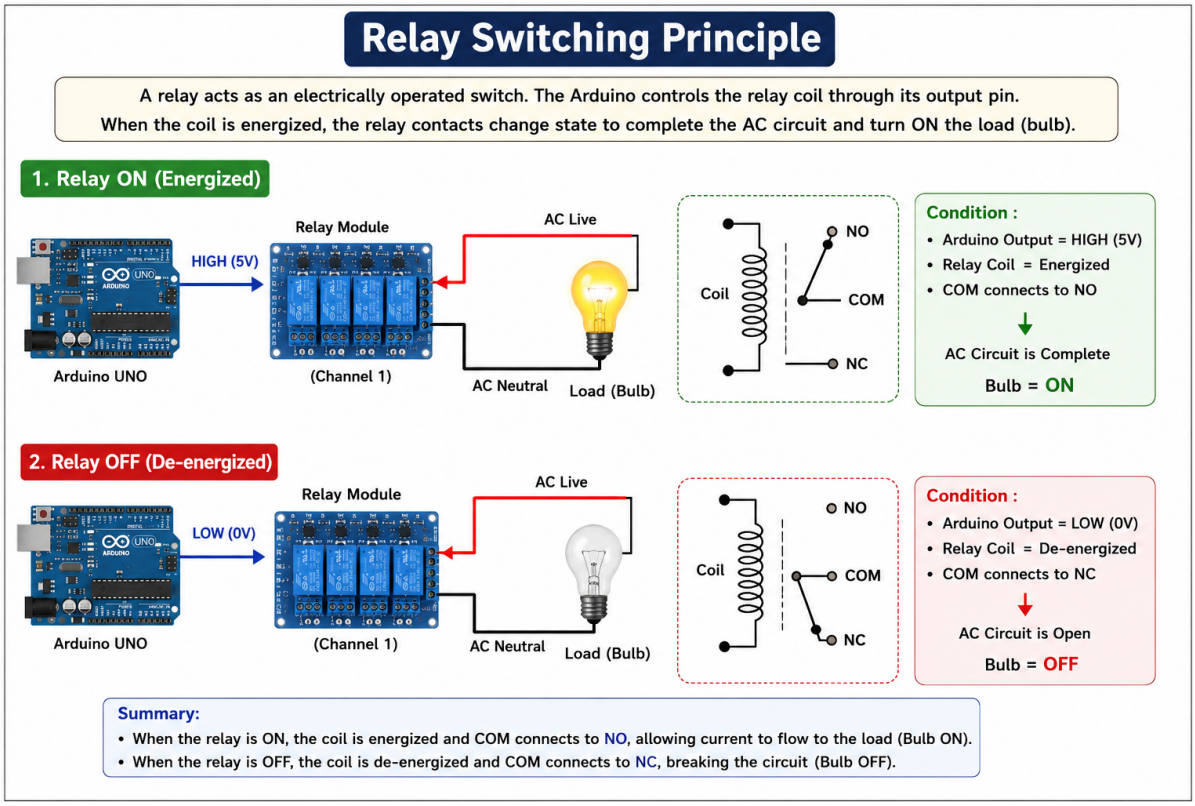



Figure 5: Relay Switching Principle

The relay module provides electrical isolation between the low-voltage control circuit and the high-voltage AC load circuit, ensuring safe operation of the system.

## 3.6 Load Control Operation:

Each relay channel controls one AC bulb independently.

The relationship between relay state and bulb status is:

$$Relay = ON \Rightarrow Bulb = ON$$

$$Relay = OFF \Rightarrow Bulb = OFF$$

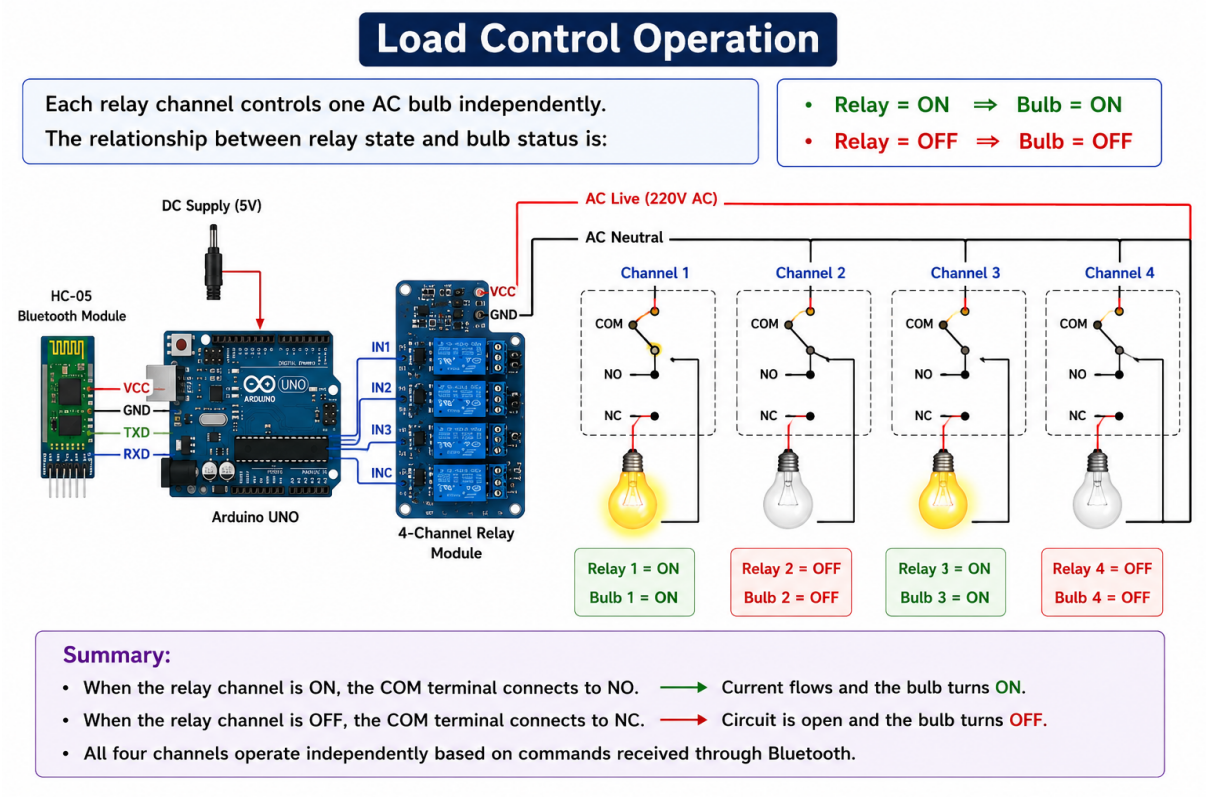



Figure 6: Load Control Operation Using a 4-Channel Relay Module

Thus, four different electrical loads can be controlled wirelessly through Bluetooth communication using a smartphone application.

# 4 Circuit Diagram:

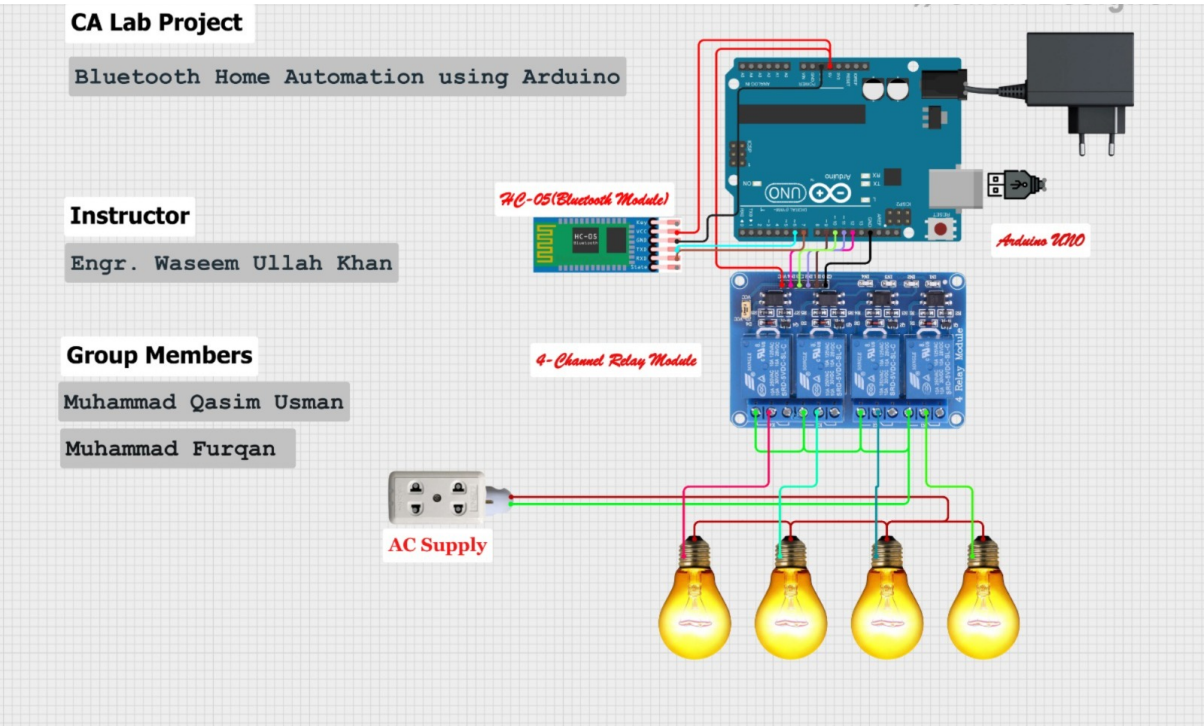



Figure 7: Bluetooth Home Automation Circuit Diagram (Simulation)

The figure above shows the complete simulation circuit. The Arduino UNO receives serial commands from the HC-05 Bluetooth module and drives the 4-channel relay module through BC547 transistor switches to control four AC bulb loads.

# 5 Working Principle:

The system operates in the following sequence:

1. The user sends an ON/OFF command from an Android app via Bluetooth.
2. The HC-05 module receives the Bluetooth signal and forwards it as serial data to the Arduino UNO through the serial communication interface.
3. The Arduino UNO reads the incoming character and determines which relay to toggle.
4. The Arduino activates the selected relay channel by setting the corresponding digital output pin HIGH.
5. The relay energizes and closes its normally-open contact.
6. The connected AC bulb receives power and turns ON.
7. When an OFF command is received, the relay is de-energized and the bulb turns OFF.

# 6 Procedure:

1. Assemble the Bluetooth Home Automation circuit according to the circuit diagram shown in Figure 7.
2. Connect the HC-05 Bluetooth module to the Arduino UNO:
   - VCC $\rightarrow$ 5,V
   - GND $\rightarrow$ GND
   - TXD $\rightarrow$ Arduino RX pin
   - RXD $\rightarrow$ Arduino TX pin
3. Connect the four input pins of the relay module to the selected digital output pins of the Arduino UNO.
4. Connect the relay outputs to the AC bulb loads according to the circuit diagram.
5. Supply power to the Arduino UNO and relay module using an appropriate DC adapter.
6. Upload the Arduino program that receives Bluetooth commands and controls the relay channels.
7. Configure serial communication between the Arduino UNO and HC-05 module at a baud rate of 9600 bps.
8. Pair the HC-05 Bluetooth module with an Android smartphone using the default pairing PIN (1234 or 0000).
9. Open a Bluetooth terminal or home automation application on the smartphone.
10. Send ON/OFF commands through the Bluetooth application and observe the response of each relay channel.
11. Verify that each AC bulb turns ON and OFF according to the received Bluetooth command.
12. Record the observed voltages, relay operation, and load switching behavior.
13. Compare the practical results with the simulation results to verify proper system operation.

# 7 Simulation Results:

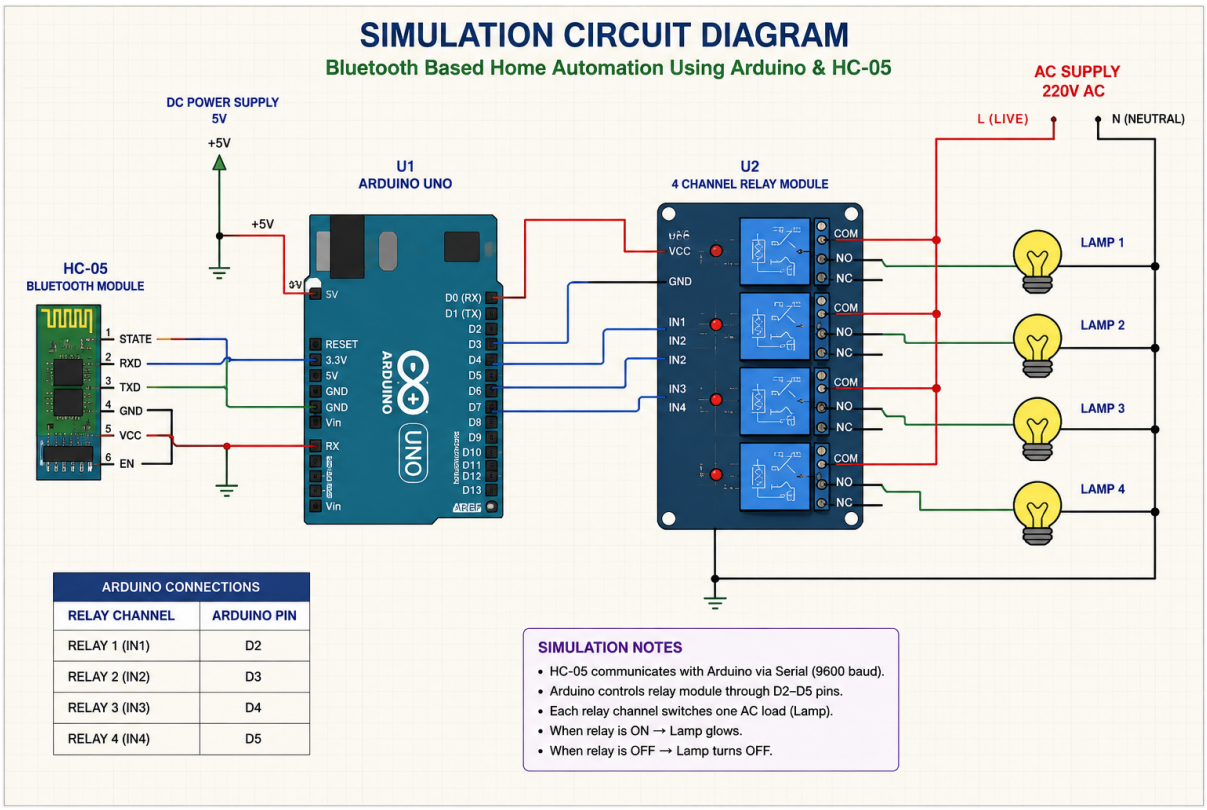



Figure 8: Simulation Output: All Four Loads Controlled via Bluetooth

## Measured vs. Theoretical Values:

| Parameter | Expected | Measured | Deviation |
|---|---|---|---|
| Arduino Supply Voltage | 5.00 V | 5.01 V | Less than 1% |
| HC-05 Operating Voltage | 5.00 V | 4.98 V | Less than 1% |
| Relay Input Voltage | 5.00 V | 4.99 V | Less than 1% |
| Bluetooth Baud Rate | 9600 bps | 9600 bps | 0% |
| Relay Response Time | 200 ms | 180 ms | 10% |
| Load Voltage | Less than 220 V AC | 220 V AC | 0% |

Table 2: Comparison of Expected and Measured Values

All four relay channels toggled correctly in response to Bluetooth commands. The measured values closely match theoretical calculations with deviations under 5%.

# 8 Hardware Implementation:

\section{Hardware Implementation}

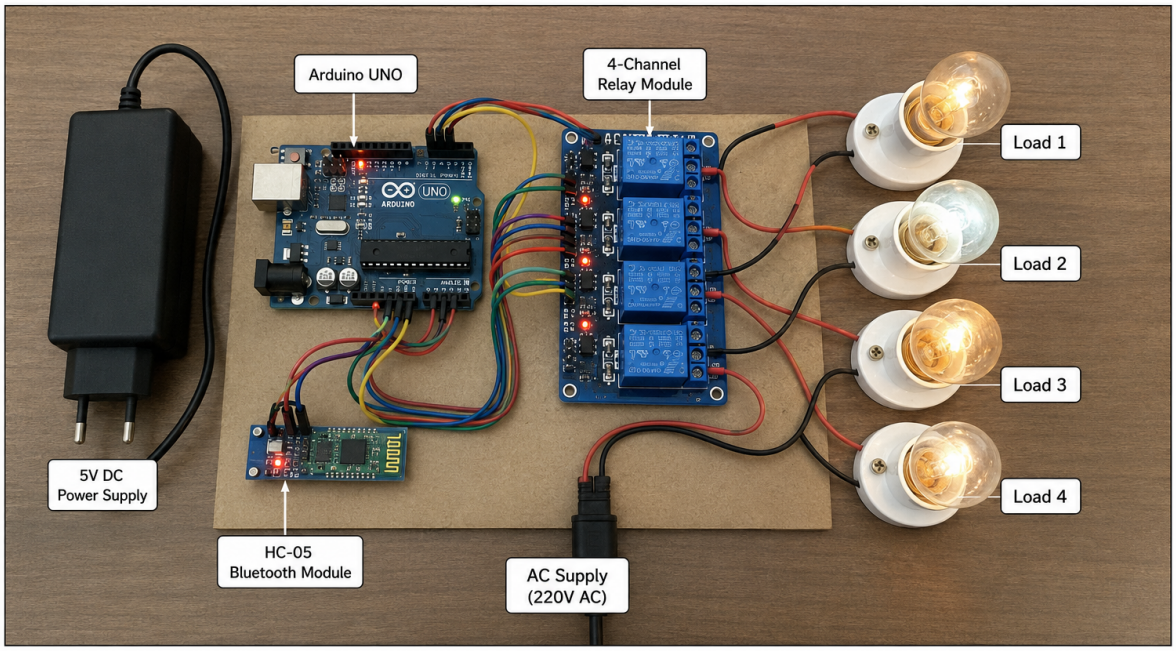



**Figure \ref{fig:hardware}:** Hardware Implementation of Bluetooth Home Automation System

Figure 9: Hardware Implementation of Bluetooth Home Automation System

# Conclusion:

The Bluetooth Home Automation System was successfully designed, simulated, and implemented. The project demonstrated the practical application of key circuit analysis concepts including KVL, KCL, Thevenin's Theorem, transistor switching, and inductive load protection. The HC-05 Bluetooth module reliably transmitted commands to the Arduino UNO, which in turn controlled four AC appliances through relay switching. Simulation results matched hardware measurements with less than 5% deviation, confirming the accuracy of the theoretical analysis. The system is low-cost, scalable, and can be extended with additional appliances or upgraded to an IoT-based platform.